

Networking Diagramming Final Project

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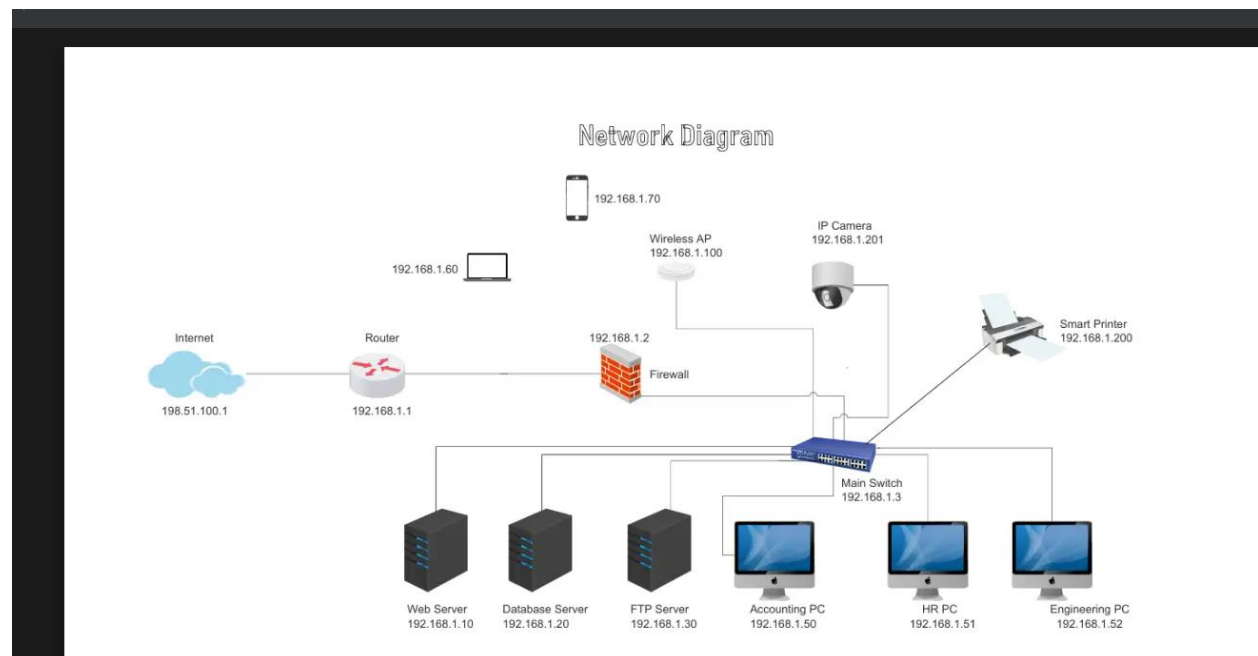
Role: Junior Penetration Tester

Scenario Summary

Acme AeroTech is a midsize aerospace parts manufacturer in the US. It specializes in the design and production of lightweight components for commercial aircrafts. It has 75 employees and a small IT team of 2 people.

The company recently received a govt contract requiring improved cyber security posture and more reliable network availability and is undergoing its first ever third party security and resilience assessment.

Initial Diagram



This is the summary of components I found in the initial diagram:

- Router
- Server
- Firewall
- Workstations
- WAP
- Switch
- Printer
- IP Camera

Vulnerabilities Discovered

On inspection of the diagram, this are the vulnerabilities I discovered:

- Lack of segmentation
 - Bad security design
 - Firewall placement
 - Server isolation
 - IP addressing inconsistencies
 - No visible access control
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- Laptop and mobile placement

Vulnerability Analysis

- **Lack of segmentation** - If a guest connects to the Wireless AP, they have a direct lateral path to sensitive data on the Accounting PC or the Database Server.
- **Bad security design** - Public-facing servers should ideally sit in a **DMZ**. As currently drawn, if the Web Server is compromised, the attacker is already inside your "trusted" internal network.
- **Firewall placement** - Web servers should sit in a Demilitarized zone to avoid intrusion.
- **Server isolation** - IoT devices are notoriously insecure and should never share a broadcast domain with sensitive servers.
- **IP addressing inconsistencies** - The Router is 192.168.1.1, the Firewall is 192.168.1.2, and the Main Switch is 192.168.1.3. While possible, usually the Firewall acts as the default gateway for the internal devices, not the router.
- **No visible access control** - Without access control intruders can make lateral movement in the network once inside.

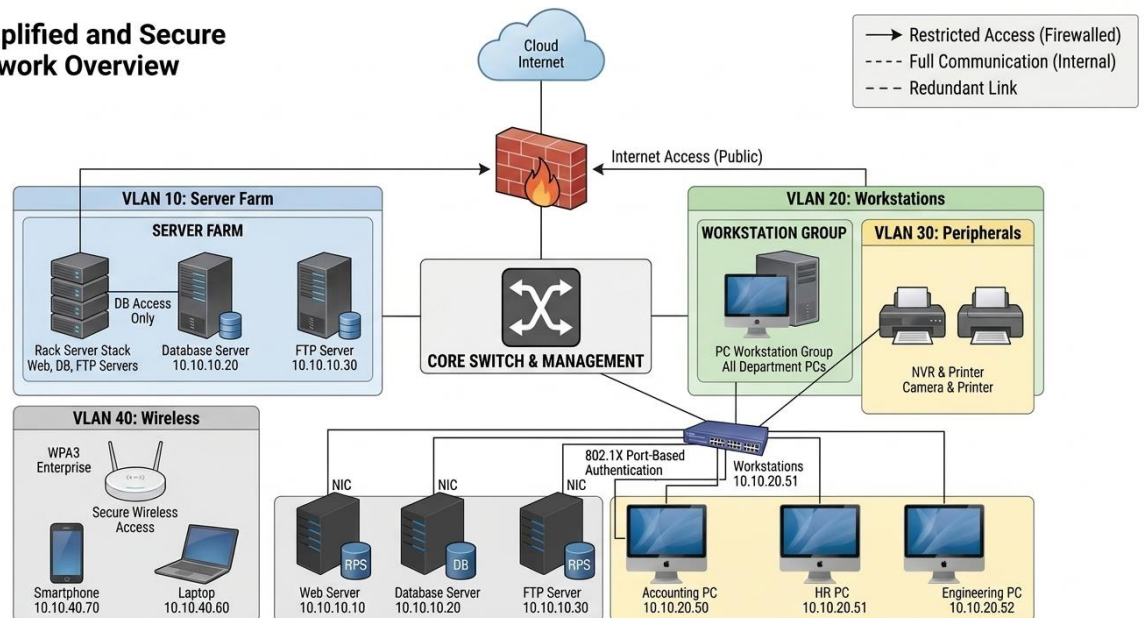
- **Laptop and Mobile placement** - The Laptop (.60) and Phone (.70) appear to be floating or connected to the Router/Firewall directly, but they likely should be communicating through the Wireless AP (.100).

Risk Prioritization

Risk	Likelihood	Impact	Priority
Lack of segmentation	high	high	high
Bad security design	high	high	high
Firewall placement	high	high	high
Server isolation	high	high	high
IP addressing inconsistencies	high	high	high
No visible access control	high	high	high
Laptop/Mobile placement	high	high	high

Redesigned Network

Simplified and Secure Network Overview



Changes Made

- **VLAN implementation** - This made the network no longer flat since it is divided into isolated zones each with its own subnet and security rules.
- **DMZ implementation** - In the first diagram, the Web Server was inside the network alongside the Database Server. All public facing servers were moved to the DMZ creating a protective barrier.
- **Firewall implementation** - We replaced the basic firewall with a modern Next-Generation Firewall to perform deep packet inspection for threats.
- **Created a dedicated peripheral network** - This separated the IoT devices from Workstations

Proposed Security Controls

- **Firewall** - We replaced the basic router with the Next-Generation Firewall that is capable of deep packet inspection that can identify and block malware.
- **Network segmentation** - Breaks the single broadcast domain into isolated zones in order to stop lateral movement during an attack.
- **Access control** - Restricts administrative access to the core switch and firewall based on users job function.
- **Endpoint hardening** - Detects and blocks ransom ware that can be brought about by insecure camera and printers.

Security Control Trade off Analysis

FACTOR	IMPACT
Cost	High Initial Investment - The upfront hardware and licensing fees are high.
Complexity	High - Managing four subnets and a DMZ is far more complex than one.
User Impact	Moderate - New devices must be registered. This will initially slowdown workflows.
Dependencies	Reliable Identity Provider - 802.1X and VPN's depend on a central directory. If your server goes down, everyone loses network access.

Timeline

Phase 1 (Week 1-2): Foundation controls

- Change default passwords
- Secure Wi-Fi(WPA3)
- Configure firewall basic rules
- Enable antivirus
- Start backups

Phase 2 (Week 3-4): Network Segmentation

- Create VLANs
- Move servers, PCs, printers into separate VLANs
- Configure inter-VLAN routing
- Apply ACLs

Phase 3 (Month 2)

- Patch all systems
- Deploy MFA for admins and remote users
- Configure monitoring/logging

Phase 4 (Month 3)

- Deploy IDS/IPS
- Introduce 802.1X pilot
- Add UPS / redundant links

Phase 5 (Ongoing)

- Monthly patching
- Quarterly access review
- Backup restore tests
- Security awareness training

Lessons Learned/Key Takeaways

- The project helped me discover the importance of having a secure network in an organization. Having a not secure network exposes the organization to breaches which might bring its systems down.
- This process made me realize how to properly put up a network that is secure. I also learned that I'm vital to the organization's security posture.

